

Hexagonal Modular Monorepo (Ports & Adapters) Component Classification & Flows

24 FASTAPI ROUTERS
[INTERNAL]

21 DOMAIN PACKAGES
[INTERNAL]

PAPER EXECUTION ADAPTER
[SIMULATED]

WORKER COORDINATOR [DISABLED BY
DEFAULT]

1. COMPONENT CLASSIFICATION & OPERATIONAL STATUS

Component Name	Subsystem Location	Classification	Implementation Status & Constraints
React Dashboard SPA	apps/dashboard/	INTERNAL	Complete React 18, Vite & TypeScript SPA across 20 distinct client routes.
FastAPI Trading Engine	apps/trading-engine/	INTERNAL	ASGI web application factory assembling 24 modular routers and Pydantic v2 schemas.
Application Services	src/services/	INTERNAL	Orchestrates orders, backtesting, walk-forward analysis, deployments & quotas.
21 Domain Packages	libraries/domain/	INTERNAL	Pure domain logic with zero database/web imports; portable quantitative IP core.
SQLAlchemy Persistence	libraries/infrastructure/	INTERNAL	Async ORM repositories managing 29 database tables across 15 linear Alembic migrations.
PostgreSQL 16 Database	Relational Database	INTERNAL / HOSTED	PostgreSQL relational target (Docker container locally; Render basic-1gb in cloud).
PaperExecutionAdapter	Execution Matching	SIMULATED	Simulates spread, adverse slippage, latency & netting; stamped is_paper=True (\$0 risk).
MockMarketDataProvider	Market Data Ingestion	SIMULATED	Offline synthetic market data generator producing deterministic historical candle bars.
Redis 7 Client	Cache & Session Store	OPTIONAL	Session & rate limit caching; includes graceful in-memory degraded fallback if offline.
TwelveData Provider	Market Data Feed	EXTERNAL	External REST client implemented in source; requires operator TWELVE_DATA_API_KEY.
OandaBrokerAdapter	Broker Sandbox	EXTERNAL	External sandbox adapter connecting to OANDA Practice environment (practice only).
Autonomous Worker	Strategy Automation	DISABLED BY DESIGN	Source implemented with manual trigger (POST /worker/run-once); loop disabled.

2. MAJOR DATA FLOW & INTERFACE SEQUENCES

Flow A: Order Submission & Simulated Execution

Trader (Client) → POST /api/v1/orders/ (CreateOrderRequest + JWT)
FastAPI Router → Validates JWT & resolves TenantContext
OrderService → Resolves active paper account & dispatches to EntitlementService
EntitlementService → Verifies daily order quota for subscription tier (e.g. 2,500/day)
RiskService → Evaluates simulated margin, leverage cap (1:100 default) & account drawdown
PaperExecutionAdapter → Models spread, adverse slippage & position netting (is_paper=True)
SQLAlchemy Session → Persists OrderModel, FillModel, PositionModel & writes ORDER_CREATE audit log
FastAPI Delivery → HTTP 201 Created (OrderResponse with is_paper=True, \$0.00 capital at risk)

Flow B: Deterministic Backtesting with LeakageGuard

Quant User (Browser) → POST /api/v1/research/experiments (ExperimentRunRequest)
ResearchService → Resolves StrategyRegistry class & loads historical candles from provider
DeterministicBacktestEngine → Sequential chronological bar traversal (t[0] → t[N])
LeakageGuard → validate_slice() verifies temporal monotonicity: timestamp[i] <= timestamp[i+1]

PerformanceCalculator → Computes Sharpe, Sortino, Max Drawdown, Win Rate & Profit Factor
FastAPI Delivery → HTTP 201 Created (ExperimentSummaryResponse with cent-level identical curves)

Flow C: Walk-Forward Analysis & Robustness Scoring

Quant User (Browser) → POST /api/v1/optimization/walk-forward (WalkForwardRunRequest)
OptimizationService → Slices dataset into rolling In-Sample (IS) & Out-Of-Sample (OOS) windows
WalkForwardEngine → Optimizes on IS; evaluates best candidate on unseen OOS window
RobustnessAnalyzer → Calculates $WFE = (Annualized\ OOS\ Return / Annualized\ IS\ Return)$
FastAPI Delivery → HTTP 201 Created (Classifies verdict: ROBUST, MODERATE, or OVERFITTED)

3. MULTI-TENANT QUERY BINDING

Every database query across accounts, orders, positions, trades, strategies, and deployments executes within a dependency-injected session context enforcing **WHERE organization_id = :org_id**. Cross-tenant data leakage is mitigated at the ORM layer.

4. PRECISION MATH & SAFETY INVARIANTS

All balances, lots, pip values, execution prices, and P&L measurements strictly use Python **Decimal** arithmetic. The execution layer contains zero code pathways to live brokerage clearing APIs. Live financial capital at risk is strictly **\$0.00**.